

Customer Shopping Behaviour Analysis

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1. Project Overview

- This Customer Shopping Behaviour Analysis examines transactional data from 3,900 purchases to understand customer segments, product preferences, and loyalty patterns, with the goal of identifying opportunities to grow revenue and improve retention.
- Key findings include a significant gap in subscription and promo code usage among female customers, a small set of age-gender segments driving the majority of revenue, and Clothing and Accessories emerging as the strongest-performing categories.
- The recommendations below outline how these insights can be translated into targeted business action.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (age, gender, location, subscription status)
 - Purchase details (item purchased, category, purchase amount, season, size, color)
 - Shopping behavior (discount applied, promo code used, previous purchases, frequency of purchases, review rating, shipping type)
- Missing Data: 37 values in the review_rating column (imputed using the mean rating — see Python section for details)

3. Exploratory Data Analysis using Python

We started with data preparation and data cleaning:

	customer_id	age	gender	item_purchased	category	purchase_amount_usd	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	promo_code_used	previous_purchases	payment_method	frequency_of_purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

- Data Importing – Imported the raw data using pandas.
- Column Cardinality – Checked the cardinality of every column in the dataset to distinguish continuous and categorical columns.

- Initial Data Exploration – Explored the data using `.describe()` to spot outliers or inconsistencies, and used `.info()` to check for missing values.
- Duplicate Data – Checked for duplicate rows in the dataset.
- Standardized Columns – Renamed the column headers in snake case format for better documentation.
- Handling Missing Values – Identified 37 null values in the `review_rating` column and replaced them with the mean rating.
- Removed Redundant Columns – Verified that `promo_code_used` and `discount_applied` were redundant (confirmed zero mismatches between the two columns) and dropped the `discount_applied` column.
- Feature Engineering –
 - Created `purchase_frequency_days` from `frequency_of_purchases`.
 - Created the `age_group` column by creating bins from the `age` column.
 - Renamed `previous_purchases` to `repeat_purchase_count` for better clarity.
- Data Exporting – Exported the cleaned dataset for use in SQL analysis.

4. Data Analysis using SQL (Structured Query Language)

After cleaning the data, we proceeded to analyse it in SQL.

- Created Database – A new database named `CustomerShopping` was created, and the cleaned data exported from Python was uploaded into it.
- Initial Validation – Checked the data for any missing rows or columns.
- Analysed the data across four parts:

Customer Overview Analysis

- Top ranked age group and gender by revenue – Identified the age group and gender combination generating the most revenue (Average revenue per customer) for the business.

	age_group	gender	total_revenue	no_of_customers	avg_revenue_per_cust
1	61-65	Female	8009	124	64
2	26-30	Female	7799	122	63
3	21-25	Male	15206	245	62
4	51-55	Female	7123	114	62
5	51-55	Male	16101	257	62
6	16-20	Female	3616	59	61
7	36-40	Male	14760	244	60
8	46-50	Male	15038	250	60
9	31-35	Female	7220	120	60
10	41-45	Female	7654	126	60

- Top 10 subscribing age group and gender by revenue – Identified the top 10 age group and gender combinations among subscribers.

	age_group	gender	total_revenue	no_of_customers	avg_revenue_per_cust
1	21-25	Male	5955	95	62.68
2	51-55	Male	7272	117	62.15
3	36-40	Male	5780	95	60.84
4	26-30	Male	6015	99	60.76
5	46-50	Male	6735	112	60.13
6	61-65	Male	6407	107	59.88
7	41-45	Male	5649	97	58.24
8	56-60	Male	5257	91	57.77
9	31-35	Male	5584	97	57.57
10	66-70	Male	4927	88	55.99

- Average purchase frequency by subscription – Compared average purchase frequency between subscribers and non-subscribers.

	subscription_status	avg_purchase_frequency
1	Yes	89
2	No	88

Product Analysis

- Highest and least selling product category by season – Identified the highest and least selling categories by volume in each season.

	season	category	sales_volume	sales_ranking	least_rank
1	Fall	Clothing	427	1	4
2	Fall	Outerwear	88	4	4
3	Spring	Clothing	454	1	4
4	Spring	Outerwear	81	4	4
5	Summer	Clothing	408	1	4
6	Summer	Outerwear	75	4	4
7	Winter	Clothing	448	1	4
8	Winter	Outerwear	80	4	4

- Top selling item from each product category – Identified the top selling item by sales revenue within each product category.

	category	item_purchased	total_sales	sales_rank
1	Accessories	Jewelry	10010	1
2	Clothing	Blouse	10410	1
3	Footwear	Shoes	9240	1
4	Outerwear	Coat	9275	1

- Product categories with review rating below 3.75 – Identified product categories with an average review rating below 3.75.

	category	avg_rating
1	Clothing	3.72
2	Outerwear	3.75

Marketing and Promotion Analysis

- Share of promo code usage by payment method – Calculated the percentage share of promo code usage across each payment mode.

	payment_method	pct_of_promo_code_used
1	Debit Card	45.44
2	Venmo	45.11
3	Bank Transfer	42.32
4	Cash	42.24
5	Credit Card	41.73
6	PayPal	41.36

- Promo code usage by high-value customers – Analysed promo code usage among high-value customers (those spending more than the average customer).

	high_spending_cust
1	839

- Relationship between promo code usage and review rating – Examined whether customers who used promo codes gave different review ratings than those who didn't.

	promo_code_used	avg_rating
1	Yes	3.74
2	No	3.76

Customer Loyalty and Repeat Behaviour

- Top 70% age group and gender – Identified the age groups and genders contributing to 70% of total revenue.

	age_group	gender	sales_count	cumulative_sales
1	56-60	Male	7031	7031
2	51-55	Male	6874	13905
3	66-70	Male	6645	20550
4	61-65	Male	6600	27150
5	21-25	Male	6389	33539
6	46-50	Male	6283	39822
7	41-45	Male	6231	46053
8	36-40	Male	6183	52236
9	31-35	Male	6164	58400
10	26-30	Male	6155	64555
11	16-20	Male	3620	68175

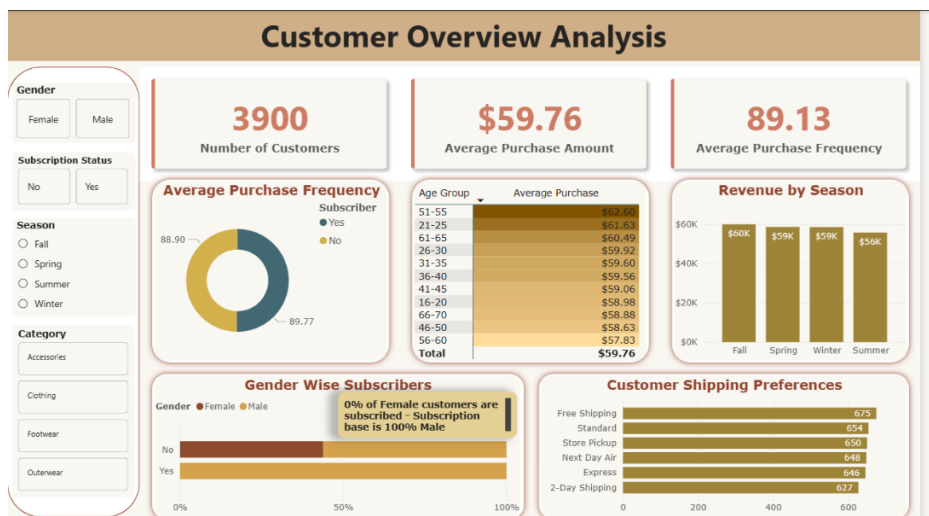
- Age groups with above-average repeat purchases – Identified age groups with repeat purchase rates above the overall average.

	age_group	avg_sales
1	56-60	26
2	66-70	26
3	51-55	26

5. Dashboard in Power BI

We built an interactive, 3-page Power BI dashboard to present these insights visually.

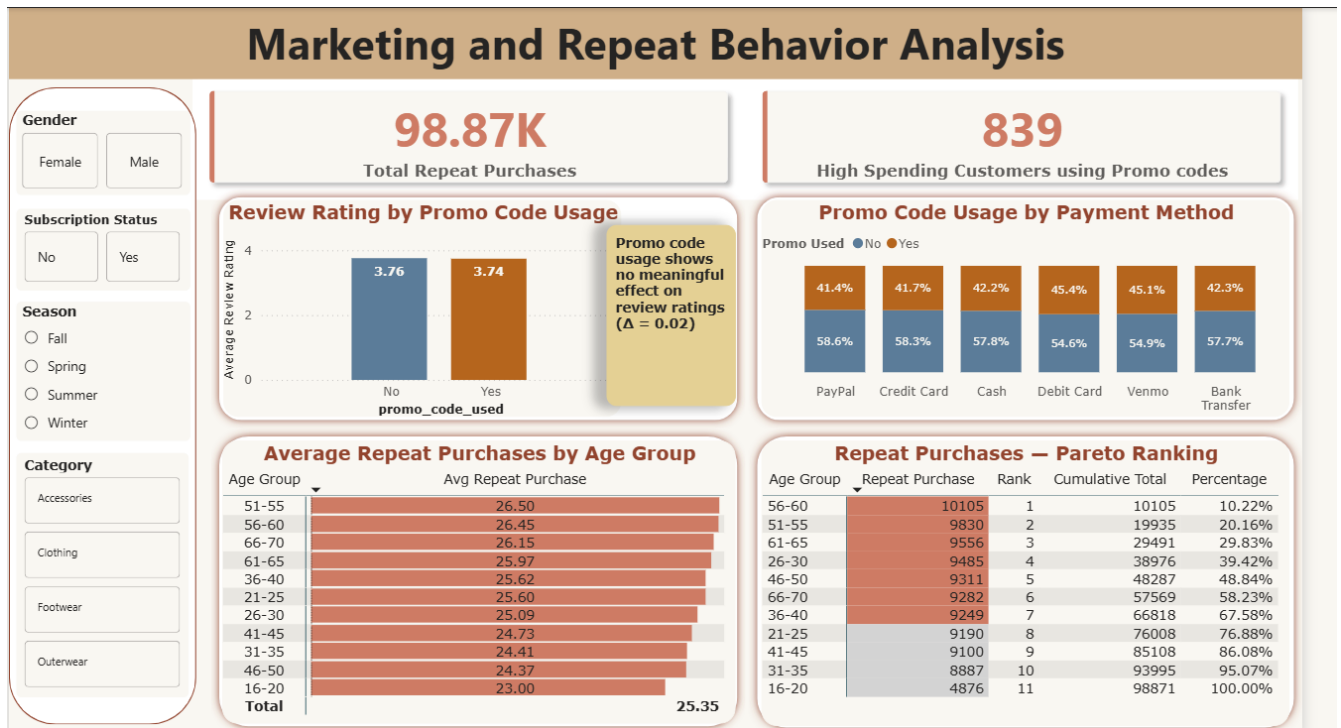
- Page 1 – Customer Overview Analysis: Presents high-level KPIs including total number of customers (3,900), average purchase amount (\$59.76), and average purchase frequency (89.13 days). Includes a breakdown of average purchase amount by age group, revenue by season, customer shipping preferences, and a donut chart comparing purchase frequency between subscribers and non-subscribers. A callout highlights that 0% of female customers are subscribed, with the entire subscription base being male.



- Page 2 – Product Analysis: Highlights revenue and performance across product categories, including total product revenue (\$233.08K) and the top revenue-generating category, Clothing (44.7%). Visuals include revenue by category, category sales volume by season, number of items offered per category, average rating by category, and a "Top Earner" table showing the best-selling item and its total sales within each category.



- Page 3 – Marketing & Repeat Behavior Analysis: Focuses on marketing effectiveness and customer loyalty, including total repeat purchases (98.87K) and the number of high-spending customers using promo codes (839). Visuals include review rating by promo code usage (highlighting a negligible 0.02 difference in ratings between users and non-users), promo code usage share by payment method, average repeat purchases by age group, and a Pareto ranking table identifying the age groups that cumulatively contribute to 70% of total repeat purchases.



All three pages share a consistent design system: a locked color palette (terracotta for KPIs and titles, teal/mustard for subscription status, amber-gold tones for category and seasonal breakdowns), synchronized slicers for Gender, Subscription Status, Season, and Category, and Edit Interactions configured to control cross-filtering behavior between specific visuals and slicers where needed. Calculated DAX measures — including ranking (RANKX), cumulative percentage, and SELECTEDVALUE-based matrix logic — power the Pareto and ranking visuals on Page 3.

6. Business Recommendations

- **Review Discount Policy** – We recommend offering more discounts, offers, and promo codes to female customers. Currently, zero female customers have used a promo code or discount, and extending these benefits could help boost sales within this segment.
- **Boost Subscriptions Among Female Customers** – We recommend reviewing the subscription scheme, as there are currently zero female subscribers. This may indicate low awareness or friction in the subscription process. A targeted campaign encouraging female customers to subscribe is recommended.
- **Increase Focus on Female Customers** – We recommend focusing more on female customers, who make up one-third of the total customer base and contribute one-third of total revenue. Expanding this segment presents a clear opportunity for business growth.
- **Optimize Shipping Preferences** – We recommend prioritizing free shipping, particularly for the Outerwear category, as nearly 20% of Outerwear's total revenue comes from orders with free shipping.
- **Review Jeans Performance** – Jeans sales are significantly lower compared to other items. We recommend reviewing the product, shipping process, and other contributing factors.
- **Purchase Frequency vs. Subscription** – The difference in average purchase frequency between subscribers and non-subscribers is minimal (only 0.16), suggesting subscription status has limited impact on purchase frequency.
- **Promo Code Usage Has Minimal Effect on Review Ratings** – Customers who used promo codes and those who didn't report nearly identical average review ratings, with a difference of only 0.02.
- **Incentivize Online Payments** – We recommend offering incentives for online payments to reduce reliance on Cash on Delivery, which would support faster cash flow.
- **Extra Focus on Clothing and Accessories** – Nearly 75% of overall revenue comes from Clothing and Accessories, with Clothing alone contributing 44.6% and Accessories 31.7%. As these are the best-performing categories, we recommend directing additional focus and resources toward both.